



Digital Electronics

Logic Gates: using RTL

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RTL Logic

- The basic circuit of the RTL digital logic family is the NOR gate.
- The voltage levels for the circuit are 0.2 V for the low level and from 1 to 3.6 V for the high level.
- The fan-out of the RTL gate is limited by the value of the output voltage when high.
- As the output is loaded with inputs of other gates, more current is consumed by the load.
- Any voltage below 1 V in the output may not drive the next transistor into saturation as required.
- The power dissipation of the RTL gate is about 12 mW.
- The propagation delay averages 25 ns.



Disadvantages of RTL Logic

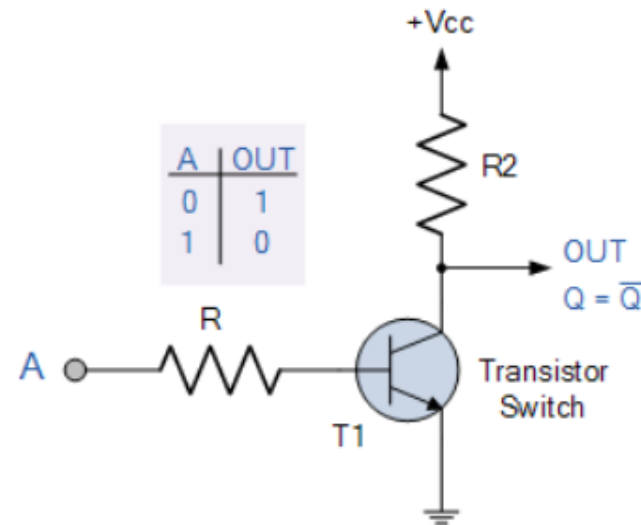
- It's relatively slow.
- Low noise immunity and noise margin.
- Low Fan-out (Approx. 3~5)
- Expensive due to fabricated of resistor.
- It can not operate above 4MHz.



NOT Gate Using RTL Logic

Transistor NOT Gate

A simple 2-input logic NOT gate can be constructed using a RTL Resistor-transistor switches as shown below with the input connected directly to the transistor base. The transistor must be saturated “ON” for an inverted output “OFF” at Q.



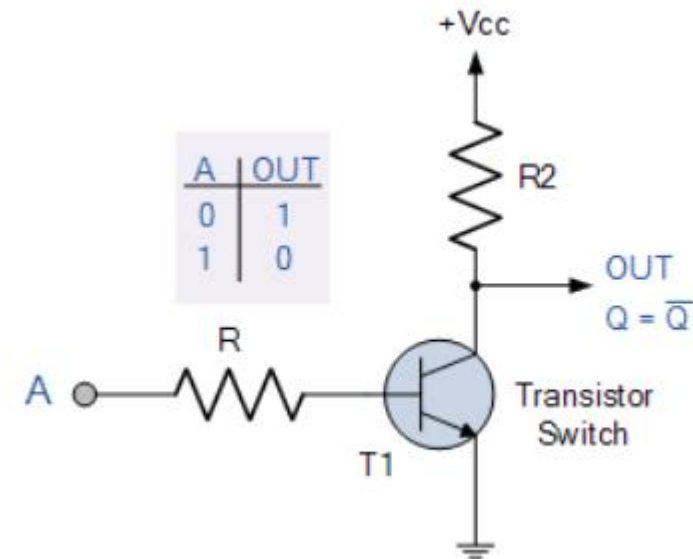


NOT Gate Using RTL Logic

Operation

Input	Transistor Status	Output
A= Low	T1=OFF , Cutoff Region	Q=+Vcc, High
A= High	T1=ON , Saturation Region	Q= $V_{CE(SAT)}=0.2\text{ V}$, Low

Circuit

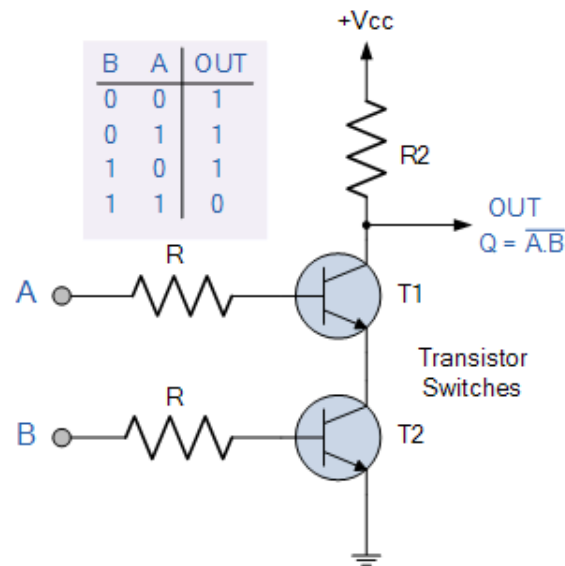




NAND Gate Using RTL Logic

Transistor NAND Gate

A simple 2-input logic NAND gate can be constructed using RTL Resistor-transistor switches connected together as shown below with the inputs connected directly to the transistor bases. Either transistor must be cut-off "OFF" for an output at Q.



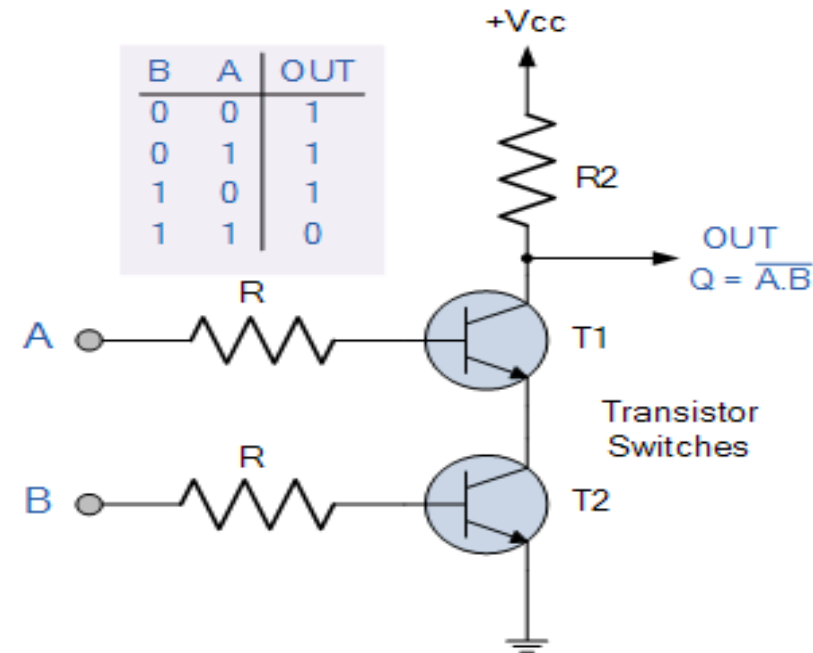


NAND Gate Using RTL Logic

Operation

Input	Transistor Status	Output
A=Low, B=Low	T1= OFF, Cutoff Region T2= OFF, Cutoff Region	Q= Vcc, High
A=Low, B=High	T1= OFF, Cutoff Region T2= ON, Saturation Region	Q= Vcc, High
A=High, B=Low	T1= ON, Saturation Region T2= OFF, Cutoff Region	Q= Vcc, High
A=High, B=High	T1= ON, Saturation Region T2= ON, Saturation Region	Q= Ground, Low

Circuit





NOR Gate Using RTL Logic

- If any input of the RTL gate is high, the corresponding transistor is driven into saturation. This causes the output to be low, regardless of the states of the other transistors.
- If all inputs are low at 0.2 V, all transistors are cut off because $V_{BE} < 0.6$ V. This causes the output of the circuit to be high, approaching the value of supply voltage V_{CC} .

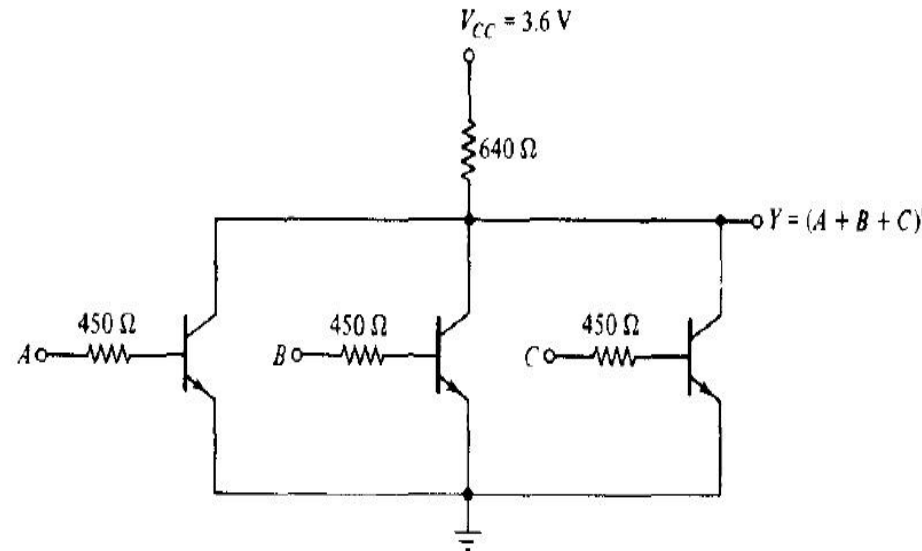


FIGURE
RTL basic NOR gate

****Related Problems will be solved in class.**



NOR Gate Using RTL Logic

Operation

Input	Transistor Status	Output
All input = low	All transistor OFF, Cutoff Region.	Output = V_{CC} , High
Any one input/Two input/all input = High	Corresponding transistor turned ON, Saturation Region	Output = $V_{CE(SAT)} = 0.2$ V. Low

Circuit

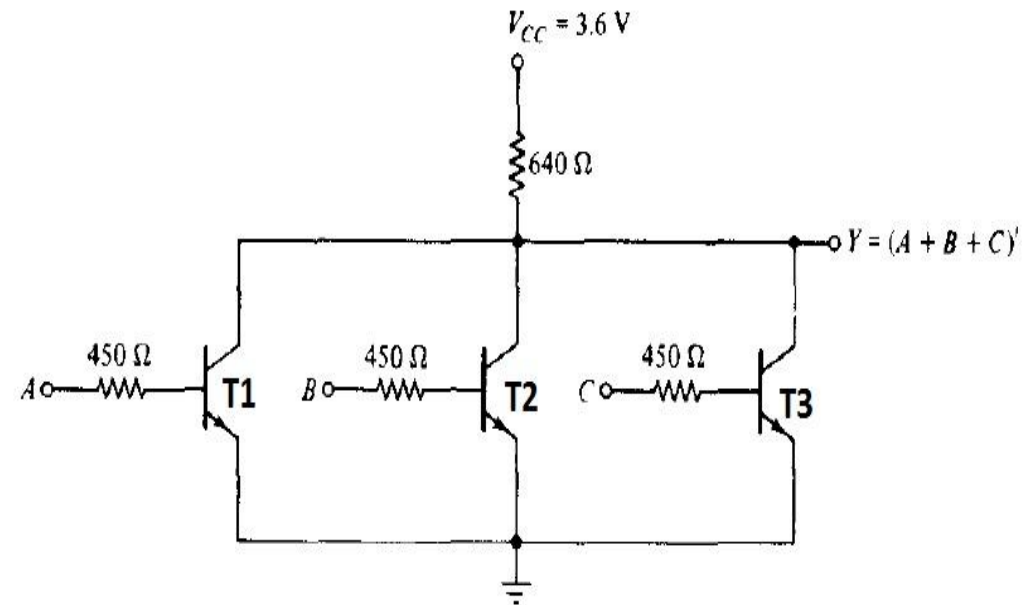


FIGURE
RTL basic NOR gate



References

1. Thomas L. Floyd, “Digital Fundamentals” 8th edition, Prentice Hall – Pearson Education.
2. M. Morris Mano, “Digital Logic & Computer Design” Prentice Hall
3. <https://www.electronics-tutorials.ws>
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